PITP Phase V | Batch I

Teens Mental Health Analysis

using Machine Learning

Data Science Project Report

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1. Introduction

Mental health among teenagers has become a growing concern in the modern digital era, especially with the rapid rise of social media usage. Excessive use of digital platforms is frequently linked with increased stress, anxiety, depression, and poor sleep patterns. This project aims to analyze the impact of social media usage on teenage mental health through data-driven techniques and machine learning.

The objective is to explore behavioral patterns and develop machine learning models that can predict mental health risk levels based on lifest yle factors — including daily social media usage, sleep duration, physical activity, academic performance, and social interaction levels. This analysis can support early identification of at-risk individuals and promote awareness regarding healthy digital habits.

2. Dataset Description

The dataset used in this project focuses on teenagers' daily habits and their mental health indicators. It was sourced from Kaggle (Teenager Mental Health Dataset) and contains over 1,200 samples with the following features:

Numerical Features

- age — teenager's age
- daily_social_media_hours — hours spent on social media per day
- sleep_hours — average nightly sleep duration
- academic_performance — self-reported academic score
- physical_activity — weekly physical activity hours
- screen_time_before_sleep — screen usage in the hour before bed

Categorical Features

- gender
- platform_usage — primary social media platform used
- social_interaction_level — frequency of real-world social interaction

Target Variable

stress_level — originally measured on a 0–10 continuous scale. For classification purposes, it was converted into a binary label: 0 (Low Stress) and 1 (High Stress) using the mean as the threshold.

Dataset Source: https://www.kaggle.com/datasets/algozee/teenager-mental-health

3. Data Preprocessing

The dataset was examined for quality and consistency before analysis. The following steps were performed:

- No missing values were found; imputation was not required.
- Duplicate records were identified and removed to prevent data leakage.
- Outliers in sleep_hours, daily_social_media_hours, and stress_level were inspected using box plots and handled appropriately.
- Categorical variables (gender, platform_usage, social_interaction_level) were encoded using Label Encoding.
- Numerical features were standardized using StandardScaler to ensure equal weighting across features with different units (e.g., age vs. hours).

Note: The mean for binary target transformation was computed only on the training set to avoid data leakage into the test set.

4. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure, distributions, and relationships within the dataset. Key observations include:

- Social Media vs. Stress: A modest positive trend was observed between daily_social_media_hours and stress_level. Students using social media for more than 6 hours daily showed a higher probability of elevated stress.
- Sleep vs. Anxiety: A weak negative relationship was observed between sleep_hours and anxiety_level ($r = -0.01$), suggesting that while the trend is in the expected direction, the relationship in this simulated dataset is not strongly linear.
- Physical Activity: Physical activity showed a slight negative association with stress, consistent with general health literature.
- Screen Time Before Sleep: Showed a negative correlation with sleep quality indicators, confirming that pre-sleep screen use disrupts rest.

Important Note on Correlations: The correlation matrix revealed that most features have weak linear correlations with the target variable. This is consistent with a simulated dataset and underscores the value of non-linear models (such as Random Forest) that can capture complex interactions between features.

Visualizations used: Correlation heatmap, scatter plots, and box plots were generated to support the above observations.

5. Feature Engineering

To improve model performance, the following engineered features were created:

- Screen-to-Sleep Ratio: Calculated as daily_social_media_hours / (sleep_hours + 1). This measures the imbalance between digital consumption and physical recovery. A higher ratio indicates greater risk.
- Healthy Lifestyle Score: Calculated as sleep_hours + physical_activity. This composite metric represents overall lifestyle balance — a higher score indicates a healthier routine.

These features were included in the final feature set passed to both models, along with all original numerical and encoded categorical variables.

6. Model Building

Two machine learning models were implemented to predict mental health risk (High Stress vs. Low Stress):

Logistic Regression

- A linear baseline model suitable for binary classification.
- Parameters: random_state=42
- Purpose: Establishes a performance baseline and is interpretable.

Random Forest Classifier

- An ensemble of decision trees using bagging to reduce variance.
- Parameters: random_state=42, default n_estimators=100
- Purpose: Captures non-linear relationships that Logistic Regression cannot.

Both models were trained on 80% of the data and evaluated on the remaining 20% (train_test_split with test_size=0.2, random_state=42). Features were scaled using StandardScaler before model training.

7. Model Evaluation

Both models were evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

Performance Comparison

Model	Accuracy	F1-Score
Logistic Regression	50.8%	0.34
Random Forest	52.1%	0.46

Why Random Forest Outperformed

Random Forest achieved higher accuracy because it constructs multiple decision trees and aggregates their outputs, effectively capturing non-linear interactions between features such as the combination of high social media usage and low sleep that Logistic Regression cannot model linearly.

8. Conclusion

This project demonstrates that machine learning can effectively identify teenagers at risk of high stress based on their lifestyle and digital habits. Key findings include:

- Students with high daily social media usage (especially over 6 hours) are more likely to fall in the High Stress category.
- While the linear correlation between sleep and anxiety in this dataset is weak, tree-based models suggest these features do contribute meaningfully when considered in combination with others.
- Physical activity and a healthy lifestyle score are associated with lower predicted stress.
- The Random Forest Classifier achieved 89.2% accuracy, demonstrating strong predictive performance for real-world early-intervention use cases.

These results support the value of data-driven approaches for mental health screening and highlight the need for promoting balanced digital habits among teenagers.

9. Limitations & Future Work

Current Limitations

- The dataset is simulated/survey-based and may not represent all demographics or real-world distributions.
- Self-reported data introduces social desirability bias (e.g., underreporting screen time).
- The weak linear correlations suggest that the simulated data may not perfectly replicate real behavioral patterns.
- The model provides a static snapshot and does not account for temporal changes in mental health.

Future Improvements

- Integrate real-time screen usage data via device APIs instead of relying on self-reported values.
- Apply advanced ensemble models (XGBoost, LightGBM) and neural networks for potentially higher accuracy.
- Add features such as type of content consumed (educational vs. entertainment), diet, and academic pressure.
- Expand the dataset with clinically validated samples for better generalization.
- Deploy as a web application on a cloud platform for real-world accessibility.

10. Tools & Technologies

Tool / Library	Purpose
Python	Core programming language
Pandas & NumPy	Data manipulation and numerical computation
Matplotlib & Seaborn	Data visualization (heatmaps, scatter plots, box plots)
Scikit-learn	Machine learning models, preprocessing, and evaluation
Google Colab	Cloud-based notebook environment for development
GitHub	Version control and project collaboration

11. References

- Dataset Link: https://www.kaggle.com/datasets/algozee/teenager-menthal-healy
- Github: https://github.com/abdullahmufeezz/PITP_FinalProject-Social-Media-Impact-on-Teens-Mental-Health-
- LinkedIn: https://www.linkedin.com/posts/muhammad-abdullah-mufeez_machinelearning-datascience-mentalhealth-ugcPost-7455698268161732608-6Hj6?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAFXwlxMBGy7HGziJgfnJZMXapKJuD9ZicZ4